# Understanding scientific pretraining ladders

A scaling ladder uses smaller training experiments to support a larger pretraining decision. This site explains Jiaxuan Zou's literature synthesis, develops the supplied technical notes, and connects practical rules to primary evidence.

## Replicate Marin, then test cyber data

Start with the PR/code/data/result lineage, stage-specific Marin evaluation, and long-context, SFT and RL experiment account. These link particular PR heads, HF revisions and result records, and distinguish public inputs from unresolved cloud dependencies.

The Datakit code walkthrough and optimizer-search manual explain the exact contracts and search procedures in pinned code. Compare Qwen3, OLMoE and Marin MoE architectures, then read the focused Step Law, Farseer and Power Lines deep dives. The cyber-data extension builds on a reproduced broad baseline.

## Start with the blog and notes

Read the Jiaxuan blog reading guide for the full section crosswalk and suggested reading order. Open the saved original blog alongside the annotated Quarto notes. The notes link their sections and subsections to detailed companions; the source review records checked claims and qualifications.

## Current industry workflow and 96-H100 plan

Start with the industry-practice audit, which checks the Jiaxuan conversation against Step Law, Farseer, Power Lines and Marin's code/records. Then read the data-engineering guide, trainer playbook, checkpoint-evaluation protocol, and 96-GPU × 30-day plan. The plan separates unique tokens from repeated exposures and includes a bounded, priced experiment queue.

The earlier 128-H100 / 40-day proposal remains a historical alternative. Its numerical model and R&D budget are not the current allocation. The SMELT review remains the architecture evidence companion.

## Start with an available corpus

The beginning-to-end pretraining design explains how to start from a Nemotron-CC snapshot or a Marin data pipeline and progress through pilots, mixture search, a scaling ladder, validation and the final training run. The OLMo and Marin deep dives explain how those teams approach each part.

## Follow a question

| Question | Explanation |
|---|---|
| What workflow should I follow? | Complete experiment workflow |
| How does a loss law choose model size and tokens? | Allocation mathematics |
| What exactly am I counting? | Parameters, FLOPs, tokens and loss |
| How should I design and fit the experiments? | Grid geometry and statistical inference |
| How do width, depth and μP work in actual ladders? | Architecture evidence and decision flowchart |

| Question | Explanation |
| --- | --- |
| How do batch, LR, decay and schedules interact? | Schedules and optimizer timescales |
| How do I choose mixtures and control repetition? | Data exposure and repetition |
| How do I predict and validate capabilities? | Task prediction and validation gates |
| What does a production recipe look like? | Nemotron pretraining case study |

The section protocols in the sidebar explain all 12 blog sections and its public ladder examples. The guides describe published practices and proposed experiment designs; this repository has not run the training experiments.

## Find and maintain the sources

Use the source housekeeping guide for the repository layout, collection status and maintenance commands. The complete source catalogue lists the papers and reports by topic with local PDFs and metadata. The Marin reading map connects reports to code examples.